MAWLANA BHASHANI SCIENCE & TECHNOLOGY UNIVERSITY

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LAB REPORT

Department of : Computer Science and Engineering (CSE)
Lab Report No : 01
Lab Report On : Images read, creating matrices and perform tasks.
Course Tittle : Data and Communication Engineering
Course Code : CSE 2210

Submitted by,	Submitted to,
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Lab 1 (1): Image read, write, display, dimension change (rgb to gray), resize.

Objective: Image read, write, display, dimension change (rgb to gray), resize operations in MATLAB.

Required Software: MATLAB

Code:

```
[file, path] = uigetfile({'*.jpg;*.png;*.bmp;*.tif', 'Image
Files (*.jpg, *.png, *.bmp, *.tif)'}, 'Select an image');

if isequal(file, 0)
    disp('Please select!');
else
    fullPath = fullfile(path, file);
    img = imread(fullPath);
    figure('Name', 'Original Image');
    imshow(img);
    title('Selected Image');

    if size(img, 3) == 3
        gray_img = rgb2gray(img);
        figure('Name', 'Grayscale Image');
        imshow(gray_img);
        title('Gray Image');
    else
        gray_img = img;
        disp('Image is already Gray');
    end

    resized_img = imresize(img, 0.5);
    figure('Name', 'Resized Image');
    imshow(resized_img);
    title('50% Resized Image');

    imwrite(gray_img, 'selected_gray_output.png');
    disp('Gray-scale image saved');
end
```

Output: The output will be as follows:

Lab 1 (2): Create two matrices manually and perform operations: add, sub, multiplication, division.

Objective: Creating two matrices manually and performing add, sub, multiplication, division operations.

Required Software: MATLAB

Code:

```
A = [1 2 3; 4 5 6];
B = [6 5 4; 3 2 1];
disp("A=")
disp(A)
disp("B=")
disp(B)
add_result = A + B;

sub_result = A - B;

mul_result = A .* B;

div_result = A ./ B;

disp("A+B=")
disp(add_result);
disp("A-B=")
disp(sub_result);
disp("A*B=")
disp(mul_result);
disp("A/B=")
disp(div_result);
```

Output: The output will be as follows:

```
A=
     1     2     3
     4     5     6

B=
     6     5     4
     3     2     1

A+B=
     7     7     7
     7     7     7

A-B=
    -5    -3    -1
     1     3     5

A*B=
     6    10    12
    12    10     6

A/B=
    0.1667    0.4000    0.7500
    1.3333    2.5000    6.0000
```

Lab 1 (3): Create two matrices by taking user input and perform add, sub, multiplication, division operations.

Objective: Creating two matrices by taking user input and performing add, sub, multiplication, division operations.

Required Software: MATLAB

Code:

```
r = input('Enter row: ');
c = input('Enter column: ');

disp('Enter Matrix A:');
A = zeros(r,c);
for i = 1:r
    for j = 1:c
        A(i,j) = input('');
    end
end

disp('Enter Matrix B:');
B = zeros(r,c);
for i = 1:r
    for j = 1:c
        B(i,j) = input('');
    end
end

disp("A=")
disp(A)
disp("B=")
disp(B)
add_result = A + B;

sub_result = A - B;

mul_result = A .* B;

div_result = A ./ B;

disp("A+B=")
disp(add_result);
```

```
disp("A-B=")
disp(sub_result);
disp("A .*B=")
disp(mul_result);
disp("A ./B=")
disp(div_result);
```

Output: The output will be as follows:

```
Enter row: 2
Enter column: 2
Enter Matrix A:
1
2
3
4
Enter Matrix B:
5
6
7
8
A=
     1     2
     3     4

B=
     5     6
     7     8

A+B=
     6     8
    10    12

A-B=
    -4    -4
    -4    -4

A .*B=
     5    12
    21    32

A ./B=
    0.2000    0.3333
    0.4286    0.5000
```